

Andrea Nigri

Associate Professor

University Address

University of Foggia
Department of Social Sciences.
andrea.nigri@unifg.it
andreanigri.wordpress.com

Permanent Address

71122 Foggia, Italy
andrea.nigri88@gmail.com

RESEARCH STATEMENT (short version)

Andrea is currently an Associate Professor in Statistics at the University of Foggia, with National Qualification as a Full Professor in Social Statistics and Demography. Prior to joining the University of Foggia, he was a Research Fellow at the Department of Social and Political Sciences, Bocconi University. He works on developing and improving the statistical methodology in the field of Longevity Risk. He has an interdisciplinary background with training in Statistics and its application to demographic research, obtained through a Ph.D. at the School of Statistics in Sapienza, University of Rome, and through the European Doctoral School of Demography, a pan-European Ph.D. training program held at the Max Planck Institute for Demographic Research and the University of Southern Denmark (Head Professor James W. Vaupel), which provided him with broad training with particular attention to mathematical and statistical demography.

Andrea has been visiting scholar at the Max-Planck Institute for Demographic Research, at the Interdisciplinary Centre on Population Dynamics, University of Southern Denmark, at the Department of Clinical Epidemiology, Aarhus University, and at Department of Actuarial Studies and Business Analytics, Macquarie University.

Prior to starting the Ph.D. program, in 2016 he was a Research Fellow at the Local Health Department of Padua ULSS6, working on epidemiologic surveillance of asbestos-related disease and occupational exposure. His research has served as a bridge between Longevity analysis and Statistical Learning, introducing Regression trees and Neural Networks as integration or full replacement of traditional mortality models.

Andrea's research has captured global attention and has been recognized by different collaborations in particular with the "Individual Finance & Insurance Decisions Centre," from which he obtained a research grant, working with Moshe A. Milevsky, Professor of mathematics and finance at York University in Toronto. Andrea has given invited academic talks in Europe, Australia, and USA, and an invited seminar by AXA that shows interest in his works. He also serves as Co-Editor-in-Chief of *Communication in Statistics. Theory and Methods*, and as an Associate Editor for different international journals in the field of Statistical Modeling, Health, and Demography.

He recently obtained funding from the Italian ministry as a co-PI "Relevant Researches of National Interest - the Italian Ministry of Education, University and Research". His current research agenda contains different strands concerning, among others, the following research questions:

- Competing risks in causes-of-death modeling.
- Age- Period- Cohort modeling for the compositional structure of causes of death.
- Multidimensional longevity modeling (Tensor decomposition approach).
- Age- Period- Cohort framework for population gap modeling. The gender gap in death rates is a prime example, using non-Gaussian distributions.

POSITION, PROFESSIONAL EXPERIENCE

**Associate Professor, University of Foggia,
Department of Social Sciences.**

Statistical Modeling.

Oct 2025 - Present

**Centre for Population Ageing Research (CEPAR) Affiliate,
University of New South Wales (UNSW Sydney, Australia),
UNSW Business School.**

Statistical Modeling.

Dec 2025 - Present

**Tenure-track Assistant Professor, University of Foggia,
Department of Economics, Management and Territory.**

Statistical Modeling.

Oct 2022 - Oct 2025

**Research fellow, Bocconi University,
Department of Social and Political Sciences**

Statistical Modeling.

Nov 2021 - Oct 2022

**Research fellow, University of Foggia,
Department of Agricultural Sciences, Food,
Natural Resources and Engineering.**

Social behavior modeling through artificial intelligence.

Nov 2020 - Oct 2021

**Research Fellow (Statistician)
at Local Health Department of Padua ULSS6.**

Epidemiological surveillance of Mesothelioma.

Sep 2016 – Aug 2017

Statistician, Kode s.r.l, Pisa.

May 2016 – Aug 2016

EDUCATION and SCIENTIFIC QUALIFICATION

National Scientific qualification as Full professor in the Italian higher education system

(Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of 13/D3 - Social Statistics and Demography. 2025 - May 2036

National Scientific qualification as associate in the Italian higher education system

(Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of 13/D1 - Statistics. 2022 - 2032

National Scientific qualification as associate in the Italian higher education system

(Ministerial Decree n. 553/2021 and 589/2021) for the disciplinary field of 13/D3 - Social Statistics and Demography. 2023 - May 2033

Ph.D in Statistics, Sapienza University of Rome,

Faculty of Statistics.

Sep 2017 - Oct 2020

Doctoral School,

International Autumn School “Demographic Methods in Public Health Research”

International Laboratory for Population and Health.

Higher School of Economics, Saint Petersburg

Sep 2018 – Oct 2018

Doctoral School,

European Doctoral School of Demography

University of Southern Denmark, Odense. (In cooperation with the Max Planck Institute for Demographic Research)

Sep

2017 – Jul 2018

Masters Degree,

Big Data.

Sapienza University of Rome,

Jan 2017 – Jan 2018

Master Degree,

Statistic for sample survey. University of Siena,

Nov 2013 – Jun 2016

Thesis: The Italian students' mobility

Bachelor of Science,

Economics. University of Foggia,

Sep 2007 – May 2013

Thesis in mathematics:

Reciprocity and Human Interactions in Game Theory.

INDIVIDUAL INTERNATIONAL RESEARCH FELLOWSHIP and GRANTS:

- **“Synoptic Assessment of Human Pressures on key Mediterranean Hot Spots.** CNR (Italian National Research Council) & Bioeconomy in Transition Research Group BiT-RG at Unitelma Sapienza - University of Rome (professor Piergiuseppe Morone). Funded by BLUEMED (BLUEMED CSA - GA 727453)” Jan 2021-Jan 2022
- **“Is Covid Gompertzian? Implications for Finance, Insurance and Economics.** Individual Finance & Insurance Decisions Centre. Professor Moshe Milevsky. May 2020- October 2020.

Founded Research Project:

- **Progetto “Dicolab. Cultura al digitale” - “1.1.6 Formazione e miglioramento delle competenze digitali” CUP F84D21000010006** (Principal Investigator) - Ricerca nei cantieri di digitalizzazione dell'investimento PNRR “Strategie e piattaforme e digitali per il patrimonio culturale” Missione 1 - Componente 3 - Misura 1- Investimento 1.1
- **Programma PE GRINS - GRINS - GROWING RESILIENT, INCLUSIVE AND SUSTAINABLE” (cod. PE00000018 CUP: H93C22000650001).** Avviso 341/2022 “Partenariati estesi alle università, ai centri di ricerca, alle aziende per il finanziamento di progetti di ricerca di base”. Piano Nazionale di Ripresa e Resilienza, Missione 4 “Istruzione e ricerca” – Componente 2 “Dalla ricerca all'impresa” – Investimento 1.3, finanziato dall'Unione europea – NextGenerationEU – BANDO a CASCATA n. 1 Progetto “Growing Resilient, INclusive and Sustainable (GRINS)”, Codice progetto PE00000018 Codice CUP H93C22000650001. Component of University of Foggia Unit. Budget: Euro 150,000.
- **Proyectos de Generación de Conocimiento, Agencia Estatal de Investigación, 2024-28: “From lifestyle behaviours to ageing: Current mortality challenges in low-mortality settings (BEHAGEING)”.** Ref: PID2023-148727OA-I00. . PIs: Sergi Trias-Llimós, Enrique Acosta. Budget: Euro 103,500.
- **PRIN PNRR 2022 Principal Investigator from 03/02/2024 and unit-P.I. before 03/02/2024 (Relevant Researches of National Interest - the Italian Ministry of Education, University and Research): Evaluation strategies to contrast university drop-out through empowerment and to design educational and career guidance practices.** University of Padua and University of Foggia. Euro 230.000
- **PRIN 2022 (Relevant Researches of National Interest - the Italian Ministry of Education, University and Research): Causes of death dependence structures and the compositional effect on overall mortality.** Bocconi (P.I. Daniele Durante), Sapienza, University of Padua. Euro 202.419
- **A data driven approach to understand and anticipate Higher Education dropout.”** Funded by Università di Foggia, 2023. Principal Investigator: Emiliano del Gobbo. finanziamento sul Fondo per Progetti di Ricerca di Ateneo

emanato con D.R. n. 1802/2022(prot. n. 55660-III/13 del 27/10/2022

- **PRIN 2017 (Relevant Researches of National Interest - the Italian Ministry of Education, University and Research): Sustaining low-impact practices in horticulture through non-destructive approach to provide more information on fresh produce history& quality.** Università di Foggia (P.I. G.Colelli)
- **Selective survival and shaping of human longevity.** Funded by Sapienza University of Rome. Principal Investigator: Elisabetta Barbi.
- **New issues in systemic risk management.** Funded by Sapienza University of Rome. Principal Investigator: Susanna Levantesi.

Research Consulting:

- **Research Consultant (Statistician)** at University of Bozen.
Monitoring of working transition. Jan 2020 – Apr 2020
- **Research Consultant (Statistician)** at University of Foggia on Creative@Hubs project.
High-dimensional data analysis with statistical and machine learning techniques. October 2022

EDITORIAL BOARDS

Permanent position

- Co-Editor in Chief: **Communications in Statistics - Theory and Methods.** (2025 -)
- Associate Editor: **Journal of Applied Statistics: Environmental Statistics and Data Science.** (2025 -).
- Associate Editor: **Journal of Population Research.** (2024 -).
- Associate Editor: **Population Health Metrics.** (2024 -).
- Editorial Board Member: **Scientific Reports.** (2023 -).
- Editorial Board Member: **Quality & Quantity.** International Journal of Methodology. (2022 -).

Guest Editor

- Guest Editor: Special Issue -Open Issues in Population Health Modeling: Methodological, and Epidemiological Implications of Continuous Changes in Population Dynamics. (2025). **Population Health Metrics**
- Guest Editor: Special Issue -Modeling and methods of population health and social change. (2024). **Scientific Reports**
- Guest Editor: Special Issue - Advances in Statistical Modelling for Social Science. (2023). **Annals of Operations Research.** *Short description: Solving practical issues, indeed, can still produce methodological improvements in the statistical domain. This special issue aims to advance the reconciliation of social science and statistics by collecting articles on cutting-edge statistical methods related to social changes, including articles on methodological developments in response to open questions in statistical modeling.*
- Lead Guest Editor: Special Issue - Methods for modelling and understanding population. (2022). changes. **Quality & Quantity.** International Journal of Methodology. *Short description: A collection of advances in statistical modeling that can enhance decision-making processes in the long and short term.*

International Research Grants Reviewer

- "European Research Council (ERC-starting grants) expert reviewer" (2025)
- "Grant proposal for the National Science Center, Poland" (2025)
- "la Caixa" Foundation (2025)
- "la Caixa" Foundation (2024)
- University of Padova (2023)
- ROCKWOOL Foundation Berlin (2023)

PUBLICATIONS

Peer Reviewed published articles.

((**A**): A-Class according to the Italian system)

- (**A**) Forecast reconciliation meets compositional data analysis: A study of Italian cause-specific age-at-death. **A. Nigri**, G.Lanfiuti Baldi, R. Mattera & HL. Shang. Journal of the Royal Statistical Society Series C: Applied Statistics. 2026. (Forthcoming)
- Spatial clustering of count differences via Skellam mixtures and Potts model. **A. Nigri**. Communications in Statistics - Theory and Methods.
- (**A**) Measuring Economic Lifespan Inequality: A New Indicator of Poverty-Free Lifespan Across Europe. **A. Nigri**. Population Health Metrics.
- (**A**) Stabilizing Inference in Dirichlet Regression via Ridge-Penalized Model. **A. Nigri**. Statistical Analysis and Data Mining. 2026. <https://doi.org/10.1002/sam.70062>
- (**A**) A New Look at the Flexible Generalized Skew-Normal Family: A Trimodal Extension and Numerical Insights. M.Bufalo & **A. Nigri**. Statistics and Computing. 2026. <https://doi.org/10.1007/s11222-026-10829-5>
- (**A**) An Age–Period–Cohort Model for the Gender Gap in Youth and Early Adult Mortality. G.Lanfiuti Baldi & **A. Nigri**. Journal of the Royal Statistical Society Series A: Statistics in Society. 2025. <https://doi.org/10.1093/jrsssa/qnaf207>
- (**A**) An Age–Period–Cohort Model in a Dirichlet Framework: A Coherent Causes of Death Estimation. R. Graziani & **A. Nigri**. Insurance Mathematics Economics (IME). 2025
- (**A**) Modeling mortality rates through hierarchical clustering for three-way data. P.Giordani, S.Levantesi, **A. Nigri**, D.Vicari. Statistics. 2025
- (**A**) The decline of ‘Deaths of Despair’ in Italy: unveiling this phenomenon in a new context. G.Lanfiuti-Baldi, **A. Nigri**, S. Trias-Llimós, E. Barbi. Population Health Metrics. 2025. <https://doi.org/10.1186/s12963-025-00430-9>
- (**A**) Extending finite mixture models with skew-normal distributions and hidden Markov models for time series. **A. Nigri**, M Forti, HL Shang. Journal of Statistical Computation and Simulation, 1-28. 2025
- (**A**) Age-period modeling of mortality gaps: the cases of cancer and circulatory diseases. GL Baldi, **A. Nigri**, HL Shang. Journal of Official Statistics. 2025
- (**A**) Stochastic variational inference for clustering short text data with finite mixtures of Dirichlet-Multinomial distributions. M Bilancia, **A. Nigri**, S Magro. Statistical Papers 66 (4), 801. 2025
- (**A**) Predicting dropout from higher education with multinomial finite mixture models: empirical evidence from Italy. M Forti, M Bilancia, B Cafarelli, E del Gobbo, **A. Nigri**. Quality & Quantity, 1-15
- (**A**) The poverty free life expectancy in Europe. G.Betti, F.Crescenzi and **A. Nigri**. Journal of Official Statistics. (2025).
- (**A**) Forecasting vital rates by a trimodal extension of the flexible generalized skew normal probability density function. M. Bufalo, **A. Nigri**. Statistics 58 (6), 1506-1530
- (**A**) Wasserstein Barycenter Regression: Application to the Joint Dynamics of Regional GDP and Life Expectancy in Italy. S.Levantesi, **A. Nigri**, P. Pagnottoni, A. Spelta. AStA Advances in Statistical Analysis
- (**A**) An interpretable cluster-based logistic regression model, with application to the characterization of response to therapy in severe eosinophilic asthma. M. Bilancia, **A. Nigri**, B. Cafarelli and D. Di Bona. The International Journal of Biostatistics. <https://doi.org/10.1515/ijb-2023-0061>
- (**A**) The Average Uneven Mortality index: Building on the ‘e-dagger’measure of lifespan inequality. M. Bonetti, U. Basellini, **A. Nigri**. Demographic Research 50, 1281-1300
- (**A**) Disaggregating death rates of age-groups using deep learning algorithms. **A. Nigri**, S. Levantesi, S. Scognamiglio. Journal of Official Statistics. (2024).
- An Evaluation of the Impact of Mitigation Policies on Health and the Economy by managing Social Distancing during Outbreaks. A. Lopolito, R. Caferra, **A. Nigri**, P. Morone. Evaluation and Program Planning. (2024)

- (A) Fuzzy clustering of the healthy life expectancy decomposition: a multi-population analysis. L. Alaimo, S. Levantesi, **A. Nigri**. Socio-Economic Planning Sciences. (2024).
- Combination of Supramicrosurgical Lymphatico-Venular Anastomosis (sLVA) and Lymph-Sparing Liposuction in Treating Cancer-Related Lymphedema: Rationale for a Regional One-Stage Approach. Gabriele, G.; **A. Nigri**, **A.**; Chisci, G.; Massarelli, O.; Cascino, F.; Komorowska-Timek, E.; Kazuki, K.; Hara, H.; Mihara, M.; Gennaro, P. J. Clin. Med. 2024, 13, 2872. <https://doi.org/10.3390/jcm13102872>
- (A) The gender gap in lifespan disparity as a social indicator of international countries: A fuzzy clustering analysis approach. L. S. Alaimo, **A. Nigri**. Socio-Economic Planning Sciences. (2023)
- (A) Lifespan variation among people with a given disease or condition. Z. Yan, I. Permanyer, V. Canudas-Romo, J.M. Aburto, **A. Nigri**, O. Plana-Ripoll. PlosOne. (2023)
- (A) A multi-way analysis of similarity patterns in longevity improvements. G. Cardillo, P. Giordani, S. Levantesi, **A. Nigri** and A. Spelta. Statistical Methods and Applications. (2023)
- (A) An application of contrast trees for mortality models diagnostic and boosting. S. Levantesi, M. Lizzi, **A. Nigri**. Quality & Quantity. (2023)
- (A) Forecasting multi-population gender gap in life expectancy. L. Cefalo, S. Levantesi, and **A. Nigri**. Social Indicators Research. (2023)
- (A) Mortality forecasting using a four-way CANDECOMP/PARAFAC decomposition. G. Cardillo, P. Giordani, S. Levantesi, **A. Nigri** and A. Spelta. Scandinavian Actuarial Journal. (2023)
- (A) Multi-country clustering-based forecasting of healthy life expectancy. S. Levantesi, **A. Nigri**, G. Piscopo, A. Spelta. Quality & Quantity, 1-27. (2023)
- (A) Evaluation of age-specific causes of death in the context of the Italian longevity transition. **Nigri, A.**, Aburto, J.M., Basellini, U., Bonetti, M. Scientific Reports 12, 22624. (2022). <https://doi.org/10.1038/s41598-022-26907-3>
- (A) A tensor-based approach to cause-of-death mortality modeling. G. Cardillo, P. Giordani, S. Levantesi, **A. Nigri**. Annals of Operations Research, 1-20. (2022)
- (A) Leveraging deep neural networks to estimate age specific mortality from life expectancy at birth. **A. Nigri**, S. Levantesi, J.M. Aburto. DEMOGRAPHIC RESEARCH. Doi: 10.4054/DemRes.2022.47.8. (2022)
- (A) A neural approach to improve the Lee-Carter mortality density forecasts. M. Marino, S. Levantesi, **A. Nigri**. North American Actuarial Journal. <https://doi.org/10.1080/10920277.2022.2050260>. (2022)
- (A) Causes-of-Death Specific Estimates from Synthetic Health Measure: A Methodological Framework. **A. Nigri**, S. Levantesi, G. Piscopo. Social Indicators Research. (2021). <https://doi.org/10.1007/s11205-021-02870-w>.
- (A) The relay for human longevity: Country-specific contributions to the increase of the best-practice life expectancy. **A. Nigri**, E. Barbi, S. Levantesi. Quality & Quantity. (2021). <https://doi.org/10.1007/s11135-021-01298-1>
- (A) Clustering-based simultaneous forecasting of life expectancy time series through Long-Short Term Memory Neural Networks. S. Levantesi, **A. Nigri**, G. Piscopo. International Journal of Approximate Reasoning. (2021) <https://doi.org/10.1016/j.ijar.2021.10.008>
- Longevity risk management through machine learning: the state of the art. S. Levantesi, **A. Nigri**, G. Piscopo. (2021) Insurance Markets and Companies
- (A) The relationship between longevity and lifespan variation. **A. Nigri**, E. Barbi, S. Levantesi. Statistical Methods & Applications. (2021). <https://doi.org/10.1007/s10260-021-00584-4>
- (A) Life expectancy and lifespan disparity forecasting: a Long Short-Term Memory approach. **A. Nigri**, S. Levantesi, M. Marino. Scandinavian Actuarial Journal. (2020) <https://doi.org/10.1080/03461238.2020.1814855>
- A random forest algorithm to improve the Lee-Carter mortality forecasting: impact on q-forward. S. Levantesi, **A. Nigri**. Soft Computing. (2019) <https://doi.org/10.1007/s00500-019-04427-z>
- A deep learning integrated Lee-Carter model. **A. Nigri**, S. Levantesi, M. Marino, S. Scognamiglio, F. Perla. Risks. (2019), 7(1):33. DOI: 10.3390/risks7010033
- Magnetic resonance imaging of the sacroiliac joints in SpA: with or without intravenous contrast media? A preliminary report. La radiologia medica. (2019). F. Gentili, L. Cantarini, M. Fabbroni, **A. Nigri**, FG. Mazzei, B. Frediani, M. Galeazzi, L. Volterrani, MA. Mazzei.

- Gastric Cancer Maximum Tumour Diameter Reduction Rate at CT Examination as a Radiological Index for Predicting Histopathological Regression after Neoadjuvant Treatment: A Multicentre GIRCG Study. Gastroenterology Research and Practice Volume. (2018). MA. Mazzei , G. Bagnacci, F. Gentili, **A. Nigri**, et al.
- High-resolution MR lymphangiography for planning lymphaticovenous anastomosis treatment: a single centre experience. La radiologia medica. (2017). MA. Mazzei, F. Gentili, FG. Mazzei, P. Gennaro, D. Guerrieri, **A. Nigri**, Gabriele G, Weber E, Fausto A, Botta G, Volterrani
- Comparison of indocyanine green fluorescence lymphography and magnetic resonance lymphangiography to investigate lymphedema of the extremities. Annali Italiani di Chirurgia. (2022) F. Gentili, F. G. Mazzei, I. Monteleone, G. Gabriele, **A. Nigri**, F. Zerini, M.G. Aboud, M.A. Mazzei, P. Gennaro.
- Covid-19 pandemic: the impact of italian lockdown on cranio-facial trauma in southern Tuscany. (2022). G. Gabriele, **A. Nigri**, B.R. Carangelo, D. Luglietto, N. Pini, F. Cascino, P. Gennaro. Annali Italiani di Chirurgia

Peer-reviewed book chapters and short papers

- Hierarchical Forecasting of Italian Mortality Data with Gender and Cause of Death Reconciliation. Giacomo Lanfiuti Baldi, Raffaele Mattera, **Andrea Nigri** & Hanlin Shang. Statistics for Innovation III Conference paper (SIS 2025).
- Exploring 'Deaths of Despair' in Italy: Pattern in Alcohol, Drug, and Suicide. Giacomo Lanfiuti Baldi, **Andrea Nigri**, Sergi Trias-Llimós. Statistics for Innovation III: SIS 2025, Short Papers, Contributed Sessions 2
- Stochastic variational inference for structured Bayesian hierarchical models. M. Bilancia, **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2024"
- Hierarchical clustering for three-way data. P. Giordani, S. Levantesi, **A. Nigri**, D. Vicari. "Book of short papers - Italian Statistical Society (SIS) 2024"
- Statistical modelling of population gaps in mortality: the Skellam distribution to model differences in death counts. G. Lanfiuti Baldi, **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2024"
- Trimodal extension based on the flexible generalized skewnormal distribution. M. Bufalo, **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2024"
- A cohort study on the gender gap in mortality through the Toker3 model. P. Giordani, S. Levantesi, **A. Nigri** and V. Zarulli. BOOK OF ABSTRACTS AND SHORT PAPERS 14th Scientific Meeting of the Classification and Data Analysis Group 2023. Published by Pearson Education Resources, Italia. ISBN: 9788891935632.
- Modelling the gender gap in youth mortality with an Age-Period-Cohort analysis. G. Lanfiuti Baldi and **A. Nigri**. Book of Short papers 11th International Conference IES 2023 Statistical Methods for Evaluation and Quality: Techniques, Technologies and Trends. ISBN 979-12-803-3369-8. <https://doi.org/10.60984/978-88-94593-36-5-IES2023>
- An Age-Period-Cohort model of gender gap in youth mortality . G. Lanfiuti Baldi and **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2023". ISBN 9788891932310
- Deep Learning framework for ungrouping coarsely aggregated vital rates Andrea Nigri. **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2022". ISBN 9788891932310
- Modelling Life Expectancy Gender Gap in a Multi-population Framework. L. Cefalo, S. Levantesi, and **A. Nigri**. MAF 2022, Mathematical and Statistical Methods for Actuarial Sciences and Finance, pp. 151–155, 2022, ISBN 978-3-030-99637-6, Springer Nature.
- An Application of the Tensor-Based Approach to Mortality Modeling. G. Cardillo, P. Giordani, S. Levantesi, and **A. Nigri**. MAF 2022, Mathematical and Statistical Methods for Actuarial Sciences and Finance, pp. 151–155, 2022, ISBN 978-3-030-99637-6, Springer Nature.
- Predicting the second wave of COVID-19 pandemic in Europe through the Dynamic Evolving Neuro Fuzzy Inference System. S. Levantesi, **A. Nigri**, G. Piscopo. Quantitative Methods in Demography: Methods and Related Applications in the Covid-19 Era (2022). eBook ISBN: 978-3-030-93005-9. Print ISBN: 978-3-030-93004-2.
- An application of contrast trees for mortality models diagnostic and boosting. S. Levantesi, M. Lizzi and **A. Nigri**. IES 2022 Innovation & Society 5.0: Statistical and Economic Methodologies for Quality Assessment BOOK OF SHORT PAPERS. ISBN 978-88-94593-36-5 online.

- The gender gap in lifespan disparity as a social indicator of international countries: A fuzzy clustering analysis approach. L. S. Alaimo, P. D'Urso and **A. Nigri**. IES 2022 Innovation & Society 5.0: Statistical and Economic Methodologies for Quality Assessment BOOK OF SHORT PAPERS. ISBN 978-88-94593-36-5 online.
- Improving longevity risk management through machine learning: a review. Chapter Book. The Essentials of Machine Learning in Finance and Accounting. S. Levantesi, **A. Nigri**, G. Piscopo.
- Estimate age-specific fertility rates from summary demographic measures. An Indirect Model Levering on Deep Neural Network. **A. Nigri**. "Book of short papers - Italian Statistical Society (SIS) 2021". ISBN 9788891927361
- Patterns in the relation between causes of death and gross domestic product. **A. Nigri** and F. Crescenzi. "Book of short papers - Italian Statistical Society (SIS) 2021". ISBN 9788891927361
- LI-CoD Model. From Lifespan Inequality to Causes of Death. **A. Nigri**, S. Levantesi. Book of short papers - Italian Statistical Society (SIS) 2020". ISBN 9788891910776

Research Consultant (Statistician)

- Dual-Energy CT for Locoregional Staging of Breast Cancer: Preliminary Results. L. Volterrani, F. Gentili, A. Fausto, V. Pelini, T. Megha, F. Sardanelli, M. A. Mazzei. AJR Am J Roentgenol. 2020 Mar;214(3):707-714. doi: 10.2214/AJR.18.20953. Epub 2020 Jan 15.
- A mortality study on male subjects exposed to polyfluoroalkyl acids with high internal dose of perfluorooctanoic acid. P. Girardi, E. Merler. Environ Res. 2019. doi: 10.1016/j.envres.2019.108743. Epub 2019 Sep 14.

INTERNATIONAL RESEARCH STAYS

- 12/11/2025 - 13/11/2025 - Short Visit at School of Risk and Actuarial Studies of the UNSW Business School. University of New South Wales (Sydney, Australia).
- 28/10/2025 - 2/12/2025 - Honorary Academic Title: Visiting Fellow at Business School in the Department of Actuarial Studies and Business Analytics. Macquarie University (Australia).
- 17/11/2024 - 27/11/2024 - Honorary Academic Title: Visiting Fellow at Business School in the Department of Actuarial Studies and Business Analytics. Macquarie University (Australia).
- 17/10/2022 - 21/10/2022 Max Planck Institute for Demographic Research. Rostock (Germany)
- 15/09/2022 - 03/10/2022 Department of Clinical Epidemiology. Aarhus University (Denmark).
- 01/09/2017 - 31/10/2017 Max Planck Institute for Demographic Research. Rostock (Germany)
- 01/11/2017 - 31/07/2018 Interdisciplinary Centre on Population Dynamics: Department of Epidemiology, Biostatistics and Biodemography. University of Southern Denmark, Odense (Denmark).

CONFERENCE ORGANIZATION and CONFERENCE BOARD MEMBER

- **Session organizer.** "Advances in Longevity Risk Modelling, Understanding, and Forecasting" at the Royal Statistical Society Conference in Edinburgh, 2025.
- **Session organizer.** "Statistical and Predictive Modeling for Social Science"; at the international conference: International Conference on Sustainability Measurement (ICSA 2024) "
- **Member of the Organizing Committee** of the international conference: International Conference on Sustainability Measurement (ICSA 2024) "
- **Session organizer.** Advances in mortality modeling at the international conference: "Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF) - Tenth edition".
- **Member of Scientific Committee** of the SMTDA 2022 and Demographics 2022 (Stochastic Modeling Techniques and Data Analysis International Conference and Demographics Workshop).
- **Member of Scientific Committee** of the SMTDA 2023 and Demographics 2023 (Stochastic Modeling Techniques and Data Analysis International Conference and Demographics Workshop).
- **Conference Chair**, Fifteenth International Longevity Risk and Capital Markets Solutions Conference. Session: Data Science-2. Washington, DC (USA) 2019.

- **Conference Chair**, Population Association of America. Session: Lifespan Inequalities around the Globe. St. Louis, (USA), 5 May 2021.

INVITED PRESENTATIONS at CONFERENCES and WORKSHOPS

- **62nd Scientific Meeting Italian Society of Economics, Demography and Statistics. Plenary Session - The Integrated One Health Approach Catania, 27–29 May 2026.** National and Subnational Trends in Longevity and Health Disparities in Italy.
- **Business School in the Department of Actuarial Studies and Business Analytics. Macquarie University (Sydney, Australia). 19, Nov. 2025.** Invited Seminar: New Looks at Cause-Specific Longevity Risk: A Distributional Approach.
- **The Australian National University. (Canberra, Australia). 11, Nov. 2025.** Invited Seminar: Mortality Gaps & Cause Composition: A Methodological View.
- **Business School in the Department of Actuarial Studies and Business Analytics. Macquarie University (Sydney, Australia). 20, Nov. 2024.** Invited Seminar: Dealing with proportions: Statistical modeling for compositional data in longevity analysis.
- **The 52nd Scientific Meeting of the Italian Statistical Society, University of Bari Aldo Moro. Bari, Italy, June 17-20, 2024.** Dealing with proportions: statistical modeling for compositional data. Solicited Sessions: Data and indicators for sustainability analysis
- **DSSR 2024 - Data Science & Social Research 4th international conference. Naples (Italy) on 25th-27th March 2024.** Invited talk: Leveraging the multiway approach for cohort gender gap analysis.
- **AIQUAV Annual Conference 2023 - Quality of life: challenges and opportunities in the crossroads of the Mediterranean. Bari (Italy) September 2023.** Invited talk: A Review on Machine Learning based frameworks for population changes modeling.
- **CLADAG 2023 - CLassification and Data Analysis Group (CLADAG) of the Italian Statistical Society (SIS). Salerno (Italy) September 2023.** Invited talk: A COHORT STUDY ON THE GENDER GAP IN MORTALITY THROUGH THE TUCKER3 MODEL.
- **SIS 2023 - Statistical Learning, Sustainability and Impact Evaluation. Ancona (Italy) June 21, 2023 – June 23, 2023.** Invited talk: Statistical learning for population studies.
- **Max Plank Institute for Demographic Research, 19 October 2022,** Invited talk: Deep learning in longevity projections. Past, present, and chance for future directions.
- **Department of Clinical Epidemiology, Aarhus University. 21, Sept. 2022 (Aarhus, DK),** Invited Seminar: Statistical modeling for mortality data. Past and future research agenda.
- **2022 Human Mortality Database (HMD) Conference. (sponsored by: AXA, Milliman, SCOR, Université Paris Dauphine), October 2022,** Invited talk: Deep Learning for Longevity projections.
- **International Conference on New Perspectives on Lifespan Inequality, July 2022 (Odense, DK),** Invited talk: Deep Learning for Longevity projections.
- **Department of Health Management & Policy at the University of New Hampshire. 12, December 2021 (Durham, NH, USA),** Invited Seminar: Leveraging deep neural networks to the indirect estimation of vital rates.
- **CPop-University of Southern Denmark. 17, April 2020 (Odense, DK),** Invited Seminar: Deep Learning in Demographic Research.
- **11th Demographic Conference of Young Demographers. 7, February 2020 (Prague, CZ),** Invited Seminar: Introduction to Machine Learning model.
- **Higher School of Economics (HSE). Special lecture at International Autumn School “Demographic Methods in Public Health Research”. September 2018 (Saint Petersburg, RU).** Invited Seminar: Machine Learning models in Demography.
- **Selected for the European Association for Population Studies Masterclass.** Prof. Jennifer Dowd (University of Oxford)

DEPARTMENTAL BOARD MEMBERSHIP

- Programme Director, Master's Degree Program in "Digital Innovation and Communication (LM-91)", Department of Social Sciences, University of Foggia. (2025-2026)
- ERASMUS Department Coordinator - Department of Economics, Management and Territory. (2024-2025)
- Student ORIENTERING Department Coordinator - Department of Economics, Management and Territory. (2024-2025)
- Member of the PhD Admission Committee - a.y. 2023-2024 - Doctoral School in Environment, Sustainability, Territory & Security; (ASTIS), University of Foggia.
- University of Foggia. Doctoral School in Environment, Sustainability, Territory & Security; (ASTIS): Member of the Doctoral Board.
- Sapienza University of Rome. Doctoral School in Statistical Sciences - Curriculum in Demography: Member of the "enlarged" Doctoral Board.
- The Seminar in Economics and Social Sciences (SEMinario in Scienze Economiche e Sociali - SEME). Department of Economics, Management and Territory, University of Foggia: Member of the Scientific Committee.

SUPERVISION

- Supervisor of Giacomo Lanfiuti Baldi, Ph.D. student at Sapienza University of Rome. Doctoral School in Statistical Sciences - Curriculum in Demography
- Supervisor of Anna Rita Ceddia, Ph.D. student at University of Foggia. ASTIS Doctoral School.
- Supervisor of Marco Forti, Postdoc at University of Foggia.

PRESENTATION

- **ORAL PRESENTATION: Royal Statistical Society International Conference 2025 (Edinburgh, UK), Sept 2025** Age-period modeling of mortality gaps: the cases of cancer and circulatory diseases, **A. Nigri**
- **POSTER PRESENTATION: Royal Statistical Society International Conference 2025 (Edinburgh, UK), Sept 2025** Extending finite mixture models with skew-normal distributions and hidden Markov models for time series, **A. Nigri**
- **ORAL PRESENTATION: Royal Statistical Society International Conference 2024 (Brighton, UK), Sept 2024** An Age-Period-Cohort Model in a DirichletFramework: A Coherent Causes of Death Estimation. R.Graziani, **A. Nigri**
- **ORAL PRESENTATION: Royal Statistical Society International Conference 2024 (Brighton, UK), Sept 2024** A methodological framework to monitor changes in proportions of Causes of Death time series.**A. Nigri**, M.Bilancia, B.Cafarelli, E.Del Gobbo.
- **POSTER PRESENTATION: Royal Statistical Society International Conference 2024 (Brighton, UK), Sept 2024** A data-driven approach to monitor and predict the careers of students at Italian universitie. M.Forti, **A. Nigri**
- **ORAL PRESENTATION: Italian Statistical Society (SIS) 2024 (Bari), June 2024** Hierarchical clustering for three-way data. P.Giordani, S.Levantesi, **A. Nigri**, D.Vicari
- **ORAL PRESENTATION: PopDays AISP 2023 (Roma), Feb 23** An Age-Period-Cohort model in a Dirichlet framework: A coherent causes of death estimation. (Nigri A., Bonetti M., Graziani R.)
- **ORAL PRESENTATION: BSPS 2022 (The British Society for Population Studies Annual Meeting), SEPT 2022.** An Age-Period-Cohort model in a Dirichlet framework: A coherent causes of death estimation. (Nigri A., Bonetti M., Graziani R.)
- **ORAL PRESENTATION: SIS 2022 (MEETING OF THE ITALIAN STATISTICAL SOCIETY), JUNE 22, 2022 – JUNE 24, 2022 (Caserta, Italy),** Deep Learning framework for ungrouping coarsely aggregated vital rates.(Nigri A.).
- **ORAL PRESENTATION: New data for the new challenges of population and society (University of Padova), SEPT 22, 2021 – SEPT 24, 2021 (Virtual),** From Life Expectancy at Birth to Vital Rates. A Novel Approach.(Nigri A., Levantesi S., Aburto JM.)

- **ORAL PRESENTATION: BSPS 2021 (The British Society for Population Studies Annual Meeting), SEPT 14, 2021 – SEPT 15, 2021 (Virtual)**, From Life Expectancy at Birth to Vital Rates. A Novel Approach.(Nigri A., Levantesi S., Aburto JM.)
- **ORAL PRESENTATION: Longevity 16 Conference. Sixteenth International Longevity Risk and Capital Markets Solutions Conference. 13th and 14th August 2021. (Copenhagen, DK)**, Indirect forecasting of cause-of-death mortality from life expectancy. (Nigri A., Levantesi S., Piscopo G.)
- **ORAL PRESENTATION: SIS 2021 (MEETING OF THE ITALIAN STATISTICAL SOCIETY), JUNE 21, 2021 – JUNE 25, 2021 (Pisa, Italy)**, Estimate age-specific fertility rates from summary demographic measures. An Indirect Model Levering on Deep Neural Network.(Nigri A.)
- **ORAL PRESENTATION: SIS 2021 (MEETING OF THE ITALIAN STATISTICAL SOCIETY), JUNE 21, 2021 – JUNE 25, 2021 (Pisa, Italy)**, Patterns in the relation between causes of death and gross domestic product. (Nigri A.)
- **ORAL PRESENTATION: PAA (Population Association of America), 10-13 April 2020 (Washington, DC (USA))**, From Life Expectancy at Birth to Vital Rates. A Novel Approach. (Nigri A.) -(Conference canceled Due to Covid-19)
- **ORAL PRESENTATION: SIS 2020 (MEETING OF THE ITALIAN STATISTICAL SOCIETY), JUNE 2020 (Pisa, Italy)**, LI-CoD Model. From Lifespan Inequality to Causes of Death. (Nigri A.) - (Conference canceled Due to Covid-19)
- **ORAL PRESENTATION: 11th Demographic Conference of Young Demographers, 5.–7. February 2020 (Prague)**, Evaluation of the age-specific causes of death, on Italian longevity improvements. (Nigri A.)
- **ORAL PRESENTATION: International Congress on Insurance: Mathematics and Economics 09/2019 (Washington, DC (USA))**, A Deep Learning Approach to Forecast Life Expectancy. (Nigri A., Levantesi S., Marino M.)
- **POSTER PRESENTATION: PAA (Population Association of America), 10-13 April 2019 (Austin, TX)**, A deep learning integrated Lee-Carter model (Nigri A., Levantesi S.).
- **ORAL PRESENTATION: PopDays AISP 2019 (Milan)**, A deep learning integrated Lee-Carter model (Nigri A., Levantesi S.).
- **POSTER PRESENTATION: PopDays AISP 2019 (Milan)**, Reciprocity Dynamics. A Dynamic Pattern to Understand an Unknown Social Behavior
- **POSTER PRESENTATION: PopFest Conference Oxford 2018**, Mortality forecasting.Machine Learning Assisted Approach (MLAA)
- **ORAL PRESENTATION: ESPAnet Italia 2017. Big Data and Social Policies. A new frontier of data management for decision-making in the social field?** Nigri A, "The Italian students' mobility".

AWARDS

- **University of Foggia Research Awards 2024.** Awards for the best scientific paper in the field of Economics - Statistical Science.

TEACHING ACTIVITY

- **Data Analysis and Data Mining (2022/23/24/25)**
Degree Programme: Communication, Public Relations and Digital Technologies
Track: Communication and Digital Technologies
- **Data Analysis and Data Mining (2022/23/24/25)**
Degree Programme: Communication, Public Relations and Digital Technologies
Track: Public Relations and Institutional Communication
- **Statistics (2025)**
Degree Programme: Economics and Management
Track: Standard track
- **Data Mining (2022/23/24/25)**
Degree Programme: Digital Innovation and Communication
Track: Standard track
- **Methodologies, Techniques and Tools for Data Analysis (2025)**
Degree Programme: Business Management and Corporate Law
Track: Standard track

PhD Courses

- **Dive into Machine and Deep Learning.** European Doctoral School of Demography (EDSD), INED Paris, June 2025
- **Intro to Statistics.** PhD in Pedagogy, University of Foggia. Fall 2024
- **Responsibility for the courses of Statistics and Machine Learning in the Doctoral School in Environment, Sustainability, Territory & Security; (ASTIS), University of Foggia.** .
- **Data Analysis.** PhD in Economics, University of Foggia. Fall 2021 (18 hours)

PhD Lectures

- **PhD Seminar University of Pisa; Department of Economics. 28, November 2019 (Pisa),** Life Expectancy and Lifespan Inequality forecasting. A Deep Learning approach.
- **PhD Seminar University of Foggia; Department of Agricultural Sciences, Food, Natural Resources and Engineering. 20, April 2021 (Foggia),** Introduction to Machine Learning models.

Other teaching

- **Beyond linear regression: An Introduction to Smoothing and Quantile Regression.** 3 hours at the Department of Statistics, Sapienza University of Rome. Master Course in Demography (Professor Barbi). (April 2020; May 2021)
- **Introduction to Machine Learning models.** 3 hours at the Department of Statistics, Sapienza University of Rome. Master Course in Demography (Professor Barbi). (April 2020; May 2021)
- **Introduction to Machine Learning models.** 6 hours at the Department of Statistics, Sapienza University of Rome. Master Course in Demography (Professor Barbi). June 2019
- **Mathematical models in Longevity.** 3 hours at the Department of Economics, Sapienza University of Rome. Master Course in Economics (Professor Bruno). June 2019

CONFERENCE

Population Association of America,	May 2021
New Perspectives on Lifespan Inequality Webinar and Open Discussion,	January 21st 2021
Statistics and Big Data at Google (Tim Hesterberg),	Padova, May 2017
Divulgare ed utilizzare l'epidemiologia dei tumori,	Padova, Dec 2016
XL Congress of the Italian Association of Epidemiology (IAE). Le evidenze in Epidemiologia: una storia lunga 40 anni,	Torino, Oct 2016

COURSES

Structural Equation Modeling (professor Claudio Barbaranelli), Sapienza University of Rome	September 2-7, 2019
3rd International Summer School on Deep Learning IRDTA (Inst. for Research Development, Training and Advice)	Warsaw, Poland - July 22-26, 2019,
Machine Learning Algorithms for Econometricians (professor Arthur Charpentier), SIde (Italian Econometric Association)	Bertinoro, 15 - 20 July 2019
Applied Bayesian Statistics School 2019 (professor Adrian Raftery), Istituto di Matematica Applicata e Tecnologie Informatiche, Consiglio Nazionale delle Ricerche	Como 24 - 28 June 2019

Deep Learning with Python for the analysis of Big Data,
Sapienza University of Rome

11 Jan.- 15 Feb. 2019

Winter School: Machine and Deep Learning for Neurological Diseases,
University of Pavia

3rd-7th Dec. 2018

Bayesian data analysis for medical data,
University of Trieste

May 2016

OTHER ACTIVITIES

- **Referee for the following journals:** International Journal of Forecasting; North American Actuarial Journal; Demography; Physica A: Statistical Mechanics and its Applications; Annals of Operation Research; Demographic Research; The Geneva Papers on Risk and Insurance; BMJ Open; Social Indicators Research; ASTIN Bulletin; Canadian Studies in Population; Genus Journal of Population Sciences; Quality & Quantity; Journal of Statistical Theory and Applications; Communications in Statistics: Case Studies, Data Analysis and Applications; British Actuarial Journal; Water journal; Sustainability journal; The Science of the Total Environment; Plos One; Decisions in Economics and Finance; Cleaner Engineering and Technology.
- **Membership:** IAML (Italian Association for Machine Learning), IBS (International Biometric Society), PAA (Population Association of America), AISP (Associazione Italiana per gli Studi di Popolazione), EPAS (European Association for Population Studies), SIDE (Italian Econometric Association), SIS (Italian Statistical Society)

SKILLS

R (Skillful),

LaTeX (Skillful),

MY SQL (Beginner),

STATA (Skillful),

SAS ENTERPRISE MINER (Beginner),

Python (Beginner),